

Boryeong Cho

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EDUCATION

KAIST – Korea Advanced Institute of Science and Technology

Integrated M.S./Ph.D. Program, Kim Jaechul Graduate School of AI
Ph.D. expected Aug 2027; Advisor: Prof. Se-Young Yun; GPA: 3.96/4.3
B.S., School of Computing; GPA: 3.86/4.3
Chungbuk Science High School

Sep 2022 – Present

Mar 2018 – Aug 2022

Mar 2015 – Feb 2018

RESEARCH INTERESTS

AI alignment · Preference optimization · Multimodal and vision-language models · Generative modeling ·
Time-series forecasting · Weather foundation models

TECHNICAL SKILLS

Programming and Tools: Python, C/C++, PyTorch, Linux

Model Development: Video transformers and diffusion, VLMs, multimodal models, preference optimization

Weather AI: HEALPix grids, lead-time conditioning, adaptive spatiotemporal resolution, NWP data

ML Systems: Multi-node distributed GPU training, data pipelines handling hundreds of TB

Algorithms: Algorithm design, data structures, heuristics, competitive programming

PUBLICATIONS

* Equal contribution; † co-corresponding authors. C: conference; J: journal; W: workshop; D: domestic conference; P: preprint.

- [C3] **Boryeong Cho**, S. Ahn[†], and S.-Y. Yun[†].
PLC-DPO: Posterior Label Correction in Noisy and Ambiguous Preference Optimization.
Findings of EMNLP, 2026 (accepted). [\[arXiv\]](#) [\[code\]](#)
- [C2] J. Kwak, N. Cao, **Boryeong Cho**, S. Lee, S. Ahn, and S.-Y. Yun.
Stage-wise Attention-Guided Region Sequencing for Adversarial Attacks on Large Vision-Language Models.
NeurIPS, 2026 (accepted). [\[arXiv\]](#)
- [C1] H. Jung*, J. Kim*, J. Kwak, **Boryeong Cho**, J. Yang, Y. Park, S. Bae, and S.-Y. Yun.
DualDrift: Combining Forward and Reverse Drifts for One-Step Generative Modeling.
NeurIPS, 2026 (accepted).
- [W3] S. Lee*, **Boryeong Cho***, H. Jung*, S. An, J. Kim, J. Kwak, Y. Yang, S. Jang, Y. Park, W. Chang, and S.-Y. Yun.
UniSAFE: A Comprehensive Benchmark for Safety Evaluation of Unified Multimodal Models.
NeurIPS 2026 Workshop on Trustworthy AI for Good (AI4GOOD) (accepted; **Oral presentation**).
[\[arXiv\]](#) [\[code\]](#) [\[dataset\]](#)
- [W2] H. Jung, J. Kim, J. Kwak, **Boryeong Cho**, J. Yang, Y. Park, S. Bae, and S.-Y. Yun.
DualDrift: Combining Forward and Reverse Drifts for One-Step Generative Modeling.
ICML Workshop on Structured Probabilistic Inference & Generative Modeling, 2026 (workshop version).
- [P1] S. Kim, **Boryeong Cho**, J. Oh, S. Chung[†], and S.-Y. Yun[†].
Rethinking Personalized Reward Modeling for LLMs under Preference Heterogeneity via Group-Debiased Federated Learning.
arXiv preprint arXiv:2608.01556, 2026. [\[arXiv\]](#)
- [J1] S. Kim*, **Boryeong Cho***, S. Bae, S. Ahn[†], and S.-Y. Yun[†].
VSCoDe: Visual-Augmentation Selection for Contrastive Decoding.
Transactions on Machine Learning Research (TMLR), 2025. [\[journal\]](#)
- [W1] S. Kim*, **Boryeong Cho***, S. Bae, S. Ahn[†], and S.-Y. Yun[†].
VACoDe: Visual Augmented Contrastive Decoding.
ICML Workshop on Trustworthy Multi-modal Foundation Models and AI Agents (TiFA), 2024.
[\[arXiv\]](#)

- [D1] M. Jeong, N. Ho, K. Park, **Boryeong Cho**, F. den Breejen, and S.-Y. Yun.
Effect of Variable Selection on Deep Learning-Based Precipitation Forecast Correction Using Large-Scale Numerical Weather Prediction Data (title translated from Korean).
Korea Software Congress (KSC), 2023.

COMPETITIVE PROGRAMMING ACHIEVEMENTS

- ICPC Asia Seoul Regional Contest: Bronze Prize, 9th place (2019); Encouragement Prize, 11th place (2020)
- Samsung Collegiate Programming Cup (SCPC): 3rd Prize, unofficial qualification (2016); 5th Prize (2020, 2022); Final Round (2018, 2021)
- Korean IOI National Team Selection Contest, 6th place, 2017
- Korea Olympiad in Informatics (KOI): Gold Medal, 3rd place (2016); Silver Medal, 6th place (2014), 7th place (2017), and 11th place (2015); Regional Grand Prize, 1st place (2014 – 2017)
- Korea Union of Clubs for Programming Contests (UCPC) Final: 4th Prize (2019); 5th Prize (2020)
- Innopolis University Open Olympiad in Informatics, represented Korea, 2017
- Asia-Pacific Informatics Olympiad (APIO), competed as Korea national representative candidate, 2014 – 2017
- KAIST ACM-ICPC Mock Competition: 3rd place (2020); 4th place (2019)
- Kookmin University Algorithm Contest, Bronze Prize (3rd place), 2017

RESEARCH PROJECTS

Yeobi: Weather Foundation Model for Global and Regional Forecasting

Role: Student Project Lead · Faculty Lead: Prof. Se-Young Yun (KAIST)

Jan 2026 – Present

Collaborators: National Institute of Meteorological Sciences (NIMS)

- Lead Yeobi development: HEALPix grids, adaptive spatiotemporal resolution, and variable attention.
- Train models on multi-node GPU clusters; build pipelines handling hundreds of TB of meteorological data.

AlphaRegion: High-Resolution Forecasting for East Asia

Role: Student Project Lead · Faculty Lead: Prof. Juho Lee (KAIST)

2025 – Present

Collaborators: NIMS and Korea Meteorological Administration (KMA)

- Lead AlphaRegion development for high-resolution forecasting over East Asia.
- Use AlphaCast global forecasts to increase both spatial and temporal resolution.

AlphaWeather: Development of Artificial Intelligence-Based Precipitation Forecasting Technology

Role: Student Researcher · Faculty Lead: Prof. Se-Young Yun (KAIST)

Sep 2022 – 2025

Collaborators: NIMS

- Managed training and evaluation of numerical weather prediction (NWP)-based forecasting models.
- Developed architectures combining inputs from multiple NWP models.
- Contributed to AlphaRain, an AI-based precipitation forecasting model under consideration for operational use by the Korea Meteorological Administration (KMA).

Developing a Computational Thinking Textbook for Teachers

Role: Student Project Lead · Faculty Lead: Prof. Kyung-Yong Chwa (KAIST)

Dec 2018 – May 2021

Collaborators: KAIST Software Education Center (KAIST SEC)

- Led the student research team and selected and developed textbook topics.
- Served as lead teaching assistant for teacher training in computational thinking.

AWARDS AND HONORS

- Government Scholarship (National Scholarship), KAIST graduate studies, 2022 – Present
- Magna Cum Laude, KAIST B.S. graduation honor, 2022
- Presidential Science Scholarship, Korea Student Aid Foundation (KOSAF), 2018 – 2022
- International Olympiad in Informatics (IOI), Korea national representative candidate, 2014 – 2017

TEACHING AND MENTORING

Instructor, Programming Master Course

Chungcheongbuk-do Education Research & Information Institute (CBERI)

Aug 2018 – Present

- Teach algorithms, data structures, and computational thinking to students preparing for informatics competitions.

Mentor, Informatics Gifted Education Program

Chungcheongbuk-do Education Research & Information Institute (CBERI)

Mar 2024 – Present

- Lead a year-long intensive mentorship program in algorithms, data structures, heuristics, and AI.
- Mentees placed 2nd and 4th at the 2026 Chungbuk Olympiad in Informatics (COI); one mentee was selected for Korea's 2026 International Olympiad in Informatics (IOI) national-team training program.
- A mentee previously placed 2nd at COI with a perfect score.

Instructor, Advanced Algorithms Class

JR Coding Education Academy

Dec 2019 – Jul 2022

- Taught algorithms and data structures for competitive programming.
- Students earned one gold, two silver, and one bronze medal at the Korea Olympiad in Informatics (KOI); two former students later joined Korea's IOI national-team training program (2026).

Teaching Assistant, IOI Korea National-Team Training

Korean Institute of Information Scientists and Engineers

2018

- Led practice sessions and explained programming problems to national-team candidates.

Teaching Assistant and Tutor, KAIST

- AI505 Optimization for AI, teaching assistant
- AI603 Machine Learning Theory, teaching assistant
- MAS102 Calculus II, tutor
- PH141 Fundamental Physics I, tutor
- CS101 Introduction to Programming, tutor

Fall 2023

Spring 2023

Fall 2019, Fall 2020

Spring 2020

Spring 2021, Fall 2021, Spring 2022

LEADERSHIP AND PROFESSIONAL ACTIVITIES

Student Representative, KAIST Graduate School of AI

2023 – Present

Represent graduate students in school affairs.

Member, Samsung Software Membership

2020 – Present

Wrote articles on algorithms and AI for the [Samsung Software Membership Blog](#) under the name VennTum (Mar 2021 – Jul 2023).

PEER REVIEW

- ACL Rolling Review (ARR), May and August 2026 cycles
- International Conference on Learning Representations (ICLR), 2027
- NeurIPS 2026 workshops: Foundations of LLM Post-Training in Changing Environments (FLLMPT); Trustworthy AI for Good (AI4GOOD)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)